

Understanding Various Reporting Commands During Synthesis Using Yosys

INTRODUCTION:

This experiment demonstrates how different reporting commands in Yosys are used to analyze the synthesized design. While Yosys does not provide as many report commands as commercial tools like Synopsys Design Compiler, it offers the stat command with several options to examine cell usage, area, and technology mapping. By experimenting with these commands, students will understand how Yosys can be used to estimate gate count, area, and simple timing/power values from a Liberty file or built-in technology models.

OBJECTIVE:

After completing this experiment, you will be able to:

- Understand the purpose of reporting in synthesis flow.
- Learn the available reporting options in Yosys.
- Analyse differences in reports with and without a Liberty file.
- Interpret Yosys statistics for gate count, area, and technology usage.

PROCEDURE:

Step 1: Create the input Design file (Verilog)

1.1: Open gvim editor for design with command: `gvim counter.v`

```
module counter(clk,a, b, x, y, sel, z);  
    input clk;  
    input [7:0] a,b,x,y;  
    input sel;  
    output reg [15:0] z;  
    always @(posedge clk) begin  
        if (sel == 1'b0)  
            z = a*b;
```

```
else  
  
z = x*y;  
  
end  
  
endmodule
```

1.2: Save the file using command- :wq!

Step 2: Synthesize the design with the following commands.

```
# Read the design  
read_verilog counter.v  
# Elaborate design  
hierarchy -check -top counter  
# Run synthesis  
proc; opt; fsm; opt; memory; opt  
techmap; opt  
# Write the synthesized netlist  
write_verilog counter_4bit.v
```

Step 3 : Use & Analyze Report Command

1.stat

- Number of wires/wire bits.
- Number of public wires.
- Number of memories, processes
- Number of instantiated cells (gates, flip-flop, etc.).

```
yosys> stat  
  
13. Printing statistics.  
  
=== counter_4bit_up ===  
  
Number of wires: 25  
Number of wire bits: 121  
Number of public wires: 3  
Number of public wire bits: 6  
Number of memories: 0  
Number of memory bits: 0  
Number of processes: 0  
Number of cells: 14  
  DFFSR 4  
  INVX1 5  
  NAND2X1 1  
  NAND3X1 1  
  XNOR2X1 2  
  XOR2X1 1
```

2. stat -liberty osu018_stdcells.lib

- Cell area report

```
yosys> stat -liberty osu018_stdcells.lib
14. Printing statistics.

=== counter_4bit_up ===

Number of wires:                25
Number of wire bits:            121
Number of public wires:         3
Number of public wire bits:     6
Number of memories:             0
Number of memory bits:         0
Number of processes:            0
Number of cells:                14
    DFFSR                        4
    INVX1                        5
    NAND2X1                      1
    NAND3X1                      1
    XNOR2X1                      2
    XOR2X1                      1

Chip area for module '\counter_4bit_up': 1012.000000
```

3. stat -tech cmos

- CMOS transistor estimate

```
yosys> stat -tech cmos
15. Printing statistics.

=== counter_4bit_up ===

Number of wires:                25
Number of wire bits:            121
Number of public wires:         3
Number of public wire bits:     6
Number of memories:             0
Number of memory bits:         0
Number of processes:            0
Number of cells:                14
    DFFSR                        4
    INVX1                        5
    NAND2X1                      1
    NAND3X1                      1
    XNOR2X1                      2
    XOR2X1                      1

Estimated number of transistors: 0+
```

4. stat -tech xilinx

- Estimates equivalent FPGA resources.
- LUTs, FFs (Logic Cells).

```
yosys> stat -tech xilinx
16. Printing statistics.

=== counter_4bit_up ===

Number of wires:                25
Number of wire bits:            121
Number of public wires:         3
Number of public wire bits:     6
Number of memories:             0
Number of memory bits:         0
Number of processes:            0
Number of cells:                14
    DFFSR                        4
    INVX1                        5
    NAND2X1                      1
    NAND3X1                      1
    XNOR2X1                      2
    XOR2X1                      1

Estimated number of LCs: 0
```

OBSERVATION:

- `stat` → shows cell, wire, and net counts.
- `stat -liberty osu018_stdcells.lib` → shows area and leakage (if present in Liberty file).
- `stat -tech cmos` → shows transistor-level estimates.
- `stat -tech xilinx` → shows FPGA LUT/FF usage estimates.

OUTPUT:

- **netlist.v** – synthesized netlist.
- **Reports** – manually generated using different `stat` options

CONCLUSION:

This experiment demonstrated the synthesis flow and manual use of reporting commands in Yosys. The synthesis script produced a netlist, and then multiple `stat` options were applied to analyze design size, area, and technology mapping. By separating synthesis and reporting, students clearly understand how reporting fits into the digital design flow.